

Green Graphene FETs: Optimized for Biosensing in the Era of Smart Mobility

1. Abstract

Graphene Nanosheet Field-Effect Transistors (GNSFETs) have also proven to be potential devices for highly sensitive and low-power biosensing devices owing to their high carrier mobility, huge surface-to-volume ratio, and capability for ultra-low power operation. Classic Tunnel FET (TFET) biosensors have been shown to provide outstanding performance in biomolecule detection, but they tend to suffer from sensitivity limitation, response time, and energy efficiency. To meet these challenges, this research builds an analytical framework for the optimization and design of GNSFET-based biosensors, Integrating systematic multi-objective optimization with physical modeling.

2. Introduction and Motivation

Biosensors are vital in healthcare, environmental monitoring, and biochemical analysis by allowing rapid biomolecule detection with high specificity. GNSFETs, taking advantage of the specific characteristics of graphene, possess a great potential over standard semiconductor-based sensors. Graphene's superior electron mobility increases the signal transduction, while its extensive surface area facilitates biomolecule adsorption, allowing for ultra-sensitive detection. In addition, GNSFETs are energy-efficient by nature and hence appropriate for wearable and portable applications. The main motivation of this research is to apply traditional TFET-based biosensor modeling to GNSFETs and methodically analyze design trade-offs using multi-objective optimization

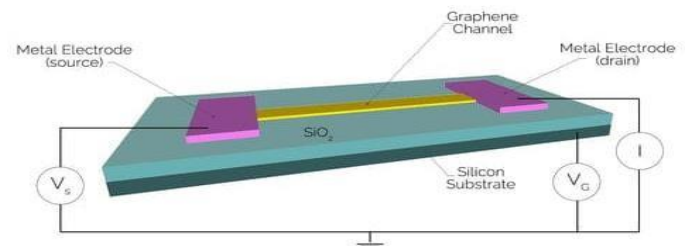


Figure 1: Graphene Nanosheet FET.

3. Background and Literature Review

Earlier research has shown TFET-based biosensors for sensing biomolecules based on surface charge modulation, and most devices trade off sensitivity, speed, and energy efficiency. The two-dimensional nature of graphene and its distinctive electronic characteristics facilitate higher sensitivity, reduced detection limits, and improved response speeds. Recent literature points to the potential of graphene-based FETs in biosensing, but few offer an integrated approach to performance optimization based on multiple conflicting objectives at once. This work fills this gap by integrating analytical modeling with an automated multi-objective optimization method.

4. Analytical Modeling of GNSFET Biosensor

The model of the proposed GNSFET biosensor involves:

- Langmuir adsorption isotherm: Describes the adsorption of biomolecules on the graphene surface, with surface coverage as a function of target concentration.
- Capacitance framework: Integrates oxide capacitance (C_{ox}) and quantum capacitance (C_q) to realistically describe charge modulation in the channel.
- Dirac point shift (ΔV_{dirac}): Accounts for the modification in the graphene channel potential by biomolecule adsorption.

Decision Variables & Performance Metrics

Performance of the GNSFET biosensor is determined by the following decision variables:

- Gate Voltage (V_g) governs channel conductivity.
- Oxide Thickness (t_{ox}) influences capacitance and sensitivity.
- Channel Length (L_{ch}) influences current and response time
- Biomolecule Concentration (C) is a measure of target analyte concentration.

Performance goals are:

1. Maximize Sensitivity
2. Minimize Response Time
3. Minimize Power Consumption
4. Minimize False Positive Rate
5. Maximize Long-Term Stability

The goals are competing by nature, requiring a multi-objective optimization approach.

Multi-Objective Optimization Approach

The NSGA-III (Non-dominated Sorting Genetic Algorithm III) is utilized to manage multiple objectives effectively. NSGA-III detects Pareto-optimal solutions with possibilities of systematic analysis of the trade-offs among sensitivity, speed, energy consumption, reliability, and stability. It provides qualitative selection of design parameters for optimum performance.

Implementation & Results

The model is coded in Python via PyCharm. Used libraries are numpy and pandas for modeling and data handling, pymoo for NSGA-III optimization, and matplotlib for plot generation. The simulation process is as follows:

1. Specifying analytical equations for the biosensor.
2. Performing NSGA-III optimization over the design space.
3. Saving output as CSV files: all solutions.csv (all the points that were run through the objective functions) and pareto_front.csv (Pareto-optimal trade-offs)..

Figure 1 : Minimize response time VS Maximize power consumption

Figure 2 : Minimize response time VS Maximize sensitivity